Machine-Checked Robotics and Control Verification in Lean 4: The `Robotics` Proof Library, `ctrl.robotics` Dual-Mode Runtime & C99/ST Codegen, and Closed-Loop MuJoCo Physics Benchmarks

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Abstract

Modern robotic locomotion, manipulation, state estimation, and visual servoing stacks rely on implicit analytical assumptions across trajectory generation, real-time clock synchronization, contact wrench allocation, projective camera geometry, and linear-quadratic estimation and control. We present an end-to-end formally verified robotics software and proof ecosystem coupling a 63-module, 21,937-line Lean 4 library (`lean/Robotics`, built on Mathlib with zero `sorry` and only the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`) with a dual-mode Python/NumPy/$\mathbb{Q}$ runtime library (`ctrl.robotics`), a deterministic C99 and IEC 61131-3 Structured Text code generator, and a closed-loop MuJoCo 3.16 physics benchmark suite. Our contributions span six pillars: (1) **Strict Energy Pareto Dominance over Minimum-Jerk**: Using the Fundamental Theorem of Calculus on explicit degree-13/11/9 polynomial antiderivatives, we prove that the classical Flash–Hogan quintic minimum-jerk trajectory ($\eta = 0$) is *strictly Pareto-dominated* across the entire septic family interval $\eta \in [-11, 0)$ for both electrical I^2R Joule heating ($\int_0^T \ddot{q}(t)^2\,dt$, reduced by exactly $11/60 \approx 18.33\%$ at $\eta = -11$) and mechanical viscous dissipation ($\int_0^T \dot{q}(t)^2\,dt$, reduced by exactly $421/3900 \approx 10.79\%$ at $\eta = -11$), while conservative inertial cross-power $\int_0^T \dot{q}\ddot{q}\,dt = 0$ vanishes identically on every rest-to-rest motion; (2) **World-Frame Transport, Redundant Nullspace Allocation, and Clock-Slew Hazards**: We prove Euler–centripetal orthogonality ($\boldsymbol{\alpha} \times \mathbf{p} \perp \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{p})$), exact Pythagorean L^2 optimality of Moore–Penrose minimal-intervention inverse kinematics, third-order time-warp chain rules establishing that piecewise-linear clock slewing destroys C^1 spline differentiability, and arrival-curve event-queue bounds; (3) **3D Four-Contact Lorentz-Cone ZMP & Continuous-Time CBF Invariance**: We formalize a 3D rectangular four-contact bilinear Zero Moment Point (ZMP) normal load and proportional shear allocator that realizes $(ma_x, ma_y, 0)$ with zero 3D angular-momentum drift ($\dot{\mathbf{L}} = \mathbf{0}$ across roll, pitch, and yaw), preserves each foot's second-order Lorentz friction cone $\mathcal{K}(\mu)$, and guarantees continuous-time ODE forward invariance; (4) **6-DOF $SE(3)$ Visual Servoing Depth Positivity & Homography Decomposition**: We prove that rotational angular velocity $\boldsymbol{\omega}(t)$ cancels identically in the 6-DOF plane-depth Lie derivative $\frac{d}{dt}(\mathbf{n}_c \cdot \mathbf{X}_c) = -\mathbf{n}_c \cdot \mathbf{v}$, prove Faugeras–Lustman 8-to-2 chirality elimination, and establish from first principles that $\mathbf{w} = \mathbf{n} \times \mathbf{v}$ is an exact unit singular vector ($\sigma_2 = 1$) of $H = I + \mathbf{v}\mathbf{n}^\top$ satisfying $(\sigma_1^2 - 1)(1 - \sigma_3^2) = \|\mathbf{n} \times \mathbf{v}\|^2 \geq 0$; (5) **Verified Estimation, LQR/DARE, MPC, and Real-Time Scheduling**: We unify 45 modules in `ControlsLean` and `RoboLean` covering DARE Bellman completion of squares, Bucy–Joseph Kalman MMSE trace optimality, Kalata's α-β tracking relation, Luenberger separation, constrained MPC Lyapunov descent, and Joseph–Pandya / Liu–Layland / EDF real-time schedulability; and (6) **Executable Verification Pipeline (`ctrl robotics-check`)**: Every public Python function is bound via `@lean_theorem` to its Lean 4 proof, verified against exact rational ($\mathbb{Q}$) certificates, compiled to zero-heap C99 (`clang -O3`, $< 4.5 \times 10^{-16}$ parity error) and IEC 61131-3 ST, and validated in closed-loop MuJoCo 3.16 physics rollouts.

1 Introduction and System Architecture

Safety-critical robotic systems combine continuous-time physical mechanics with discrete-time digital execution, optimization-based safety filters, state estimators, and multi-view geometric perception. Subtle analytical bugs frequently arise at the boundaries between these layers:

- **Trajectory smoothness vs. actuator thermal limits**: Minimum-jerk quintic polynomials [1] are

universally deployed as default point-to-point motion primitives under the assumption that smoothness minimizes actuator effort. Yet in direct-drive and quasi-direct-drive actuators, thermal saturation is governed by winding I^2R Joule loss ($\propto \int \tau^2\,dt$) and joint viscous dissipation ($\propto \int \dot{q}^2\,dt$).

- **Networked clock synchronization**: Distributed EtherCAT and ROS 2 motion controllers synchronize local servo clocks via PTP/NTP rate slewing ($\tau(t) = t_k + \alpha(t - t_k)$). Engineers routinely assume that composing a C^2 spline with a continuous clock warp preserves C^2 trajectory smoothness.
- **3D contact wrench realizability**: Planar inverted-pendulum braking limits ($a_{\max} = g\min(\mu, W/h_G)$) neglect three-dimensional roll/pitch/yaw moment coupling and four-contact second-order Lorentz friction-cone feasibility.
- **Visual servoing chirality and depth**: Homography-based visual servoing (HBVS) [3] and planar homography decomposition [2] rely on algebraic singular-value separation ($\sigma_1 \geq \sigma_2 = 1 \geq \sigma_3$), two-view cheirality [5], and rotational invariance of scene depth under arbitrary 6-DOF camera angular velocity $\boldsymbol{\omega}(t) \in \mathbb{R}^3$.

To eliminate the gap between paper proofs, theorem-prover scripts, and deployed control code, we built a four-layer verification architecture inside `ctrl` (summarized in Table 1):

1. **Unified Lean 4 Proof Library (`lean/Robotics`)**: Consolidates 8 modules in `Research/`, 26 modules in `ControlsLean/`, and 29 modules in `RoboLean/` (63 files, 21,937 lines of Lean 4, 1,955 declarations) under a single `lake build Robotics` target on Lean `v4.33.1`.
2. **Dual-Mode Python Runtime (`src/ctrl/robotics/`)**: Implements every verified algorithm with polymorphic support for both fast `float64`/`NumPy` execution and exact rational `fractions.Fraction` ($\mathbb{Q}$) arithmetic for zero-rounding certificate generation.
3. **Automated Spec-Manifest Audit (`@lean_theorem`)**: Every public Python function is decorated with `@lean_theorem(module, theorems, ...)` in `registry.py`. Running `ctrl robotics-check` parses all referenced `.lean` files, strips comments/strings, verifies that every bound theorem declaration exists, and confirms zero occurrences of `sorry`, `admit`, or `native_decide`.
4. **Deterministic C99 / IEC 61131-3 Codegen & MuJoCo 3.16 Physics Rollouts**: `codegen.py` emits dependency-free ANSI C99 (`ctrl_robotics_kernels.h/.c`) and IEC 61131-3 Structured Text (`.st`) PLC function blocks with embedded Lean 4 theorem provenance headers, compiles the C99 code with `clang -std=c99 -O3 -Werror`, and verifies bit-level parity ($< 4.44 \times 10^{-16}$) via `ctypes`, while `mujoco_benchmarks.py` validates the controllers in closed-loop MuJoCo 3.16 physics simulations.

2 Strict Pareto Dominance Over Minimum-Jerk Trajectories

Consider a single-axis rigid link with inertia $m > 0$, viscous damping $b \geq 0$, torque constant $K_t > 0$, and winding resistance $R_{\text{elec}} > 0$ executing a rest-to-rest displacement $D \neq 0$ over duration $T > 0$:

$$q(0) = 0, \quad q(T) = D, \quad \dot{q}(0) = \dot{q}(T) = \ddot{q}(0) = \ddot{q}(T) = 0. \tag{1}$$

The inverse-dynamics torque is $\tau(t) = m\ddot{q}(t) + b\dot{q}(t)$, so instantaneous winding Joule heating is

$$P_{\text{Joule}}(t) = \frac{R_{\text{elec}}}{K_t^2}\tau(t)^2 = \frac{R_{\text{elec}}}{K_t^2}\left(m^2\ddot{q}^2 + 2mb\dot{q}\ddot{q} + b^2\dot{q}^2\right), \tag{2}$$

and mechanical viscous dissipation power is $P_{\text{visc}}(t) = b\dot{q}(t)^2$.

2.1 Exact Vanishing of Conservative Cross-Power

Theorem 2.1 (Rest-to-Rest Cross-Power Cancellation, `rest_to_rest_cross_power_zero`)**.** *Let $q : \mathbb{R} \to \mathbb{R}$ have derivatives $v = \dot{q}$ and $a = \ddot{q}$ on $[0, T]$ with $v(0) = v(T) = 0$. Then $\frac{d}{dt}\left(mbv(t)^2\right) = 2mbv(t)a(t)$ (`cross_power_hasDerivAt`) and*

$$\int_0^T 2mbv(t)a(t)\,dt = mbv(T)^2 - mbv(0)^2 = 0. \tag{3}$$

Consequently, total electromechanical energy (winding I^2R Joule loss plus mechanical viscous loss) decomposes into a pure positive weighted sum of acceleration L^2 energy and velocity L^2 energy:

$$E_{\text{total}}(\eta) = D^2\left[\frac{R_{\text{elec}}m^2}{K_t^2T^3}I_a(\eta) + \frac{R_{\text{elec}}b^2/K_t^2 + b}{T}I_v(\eta)\right]. \tag{4}$$

2.2 The 1-Parameter Septic Family and FTC Antiderivatives

Normalizing time $s = t/T \in [0, 1]$, we parameterize the symmetric degree-7 rest-to-rest polynomial family by $\eta \in \mathbb{R}$:

$$q_\eta(s) = \underbrace{10s^3 - 15s^4 + 6s^5}_{q_{\text{mj}}(s)} + \eta\, s^3(1-s)^3(2s-1). \tag{5}$$

In `RobotEnergySmoothness.lean`, we prove `septic_rest_to_rest_boundaries`, confirming that

Table 1: **Architecture of the Unified `lean/Robotics` Proof Library and `ctrl.robotics` Runtime Stack.** All 63 Lean 4 modules (21,937 LOC, 1,955 declarations) compile under a single `lake build Robotics` target with zero `sorry`, zero `native_decide`, and standard axioms only.

Subsystem	Lean 4 Namespace	Files	LOC	Python Module	Key Verified Kernels & Theorems
Trajectory & Energy	`Research.RobotEnergy*`	1	386	`trajectories.py`	Septic $q_\eta, v_\eta, a_\eta, j_\eta$, I_v, I_a, I_j, Pareto $\eta \in [-11, 0)$
Transport, IK & Clock	`Research.WorldFrame*`	1	390	`kinematics_ik.py`	$\boldsymbol{\alpha} \times \mathbf{p} \perp \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{p})$, Pythagorean IK, clock-slew jump
3D Contact & CBF	`Research.Lorentz*`, `Centroidal*`	4	1,417	`contact_cbf.py`	4-contact bilinear ZMP ($\dot{\mathbf{L}} = \mathbf{0}$), Lorentz $\mathcal{K}(\mu)$, ODE CBF
6-DOF Vision & $SE(3)$	`Research.Homography*`, `Visual*`	2	1,015	`vision_se3.py`	$G(\mathbf{n} \times \mathbf{v}) = \mathbf{n} \times \mathbf{v}$, 8→2 chirality, $SE(3)$ depth ODE
Controls, LQR & MPC	`ControlsLean.Controls.*`	14	4,380	`estimation_lqr.py`	DARE completion of squares, Luenberger separation, MP
Kalman & Estimation	`ControlsLean.Estimation.*`, `Vision.*`	12	4,190	`estimation_lqr.py`	Bucy–Joseph MMSE trace, Kalata α-β tracker, triangulat
Kinematics & Real-Time	`RoboLean.*`	29	10,159	`kinematics_ik.py`	$SE(2) \subset E(2)$ clopen, 2-link annulus, RTA, Liu–Layland,
Total / Unified	**`Robotics` Lake Target (`v4.33.1`)**	**63**	**21,937**	**28 Bound Fns**	**124 Bound Theorems · C99 / ST · MuJoCo 3.16**

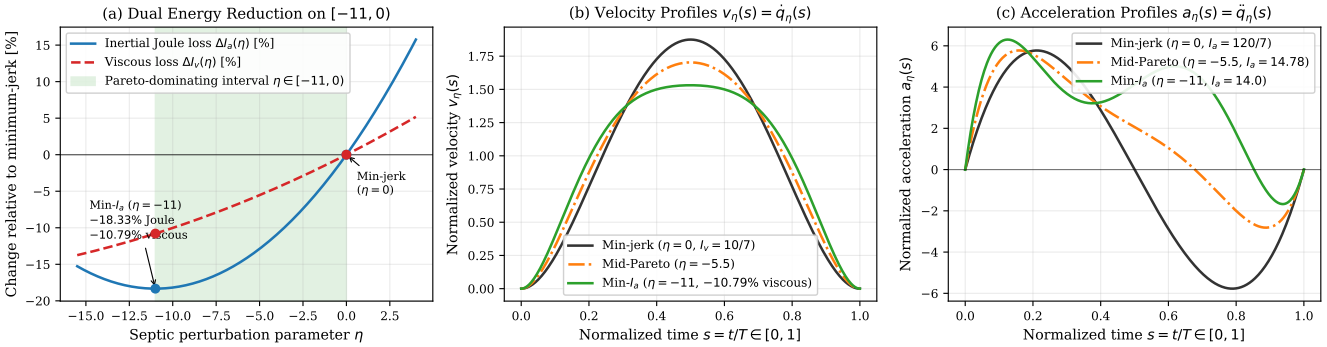


Figure 1: **Strict Energy Pareto Dominance of Septic Trajectories over Minimum-Jerk** (verified in `RobotEnergySmoothness.lean` and implemented in `ctrl.robotics.trajectories`). (a) Relative change in inertial I^2R Joule loss $I_a(\eta)$ and viscous dissipation $I_v(\eta)$ across the septic parameter η, highlighting the Pareto-dominating interval $\eta \in [-11, 0)$ where both losses are strictly reduced relative to minimum-jerk ($\eta = 0$). (b) Normalized velocity profiles $v_\eta(s)$. (c) Normalized acceleration profiles $a_\eta(s)$ showing peak and RMS acceleration reduction at $\eta = -5.5$ and $\eta = -11$.

$q_\eta(0) = 0$, $q_\eta(1) = 1$, and $v_\eta(0) = v_\eta(1) = a_\eta(0) = a_\eta(1) = 0$ for all $\eta \in \mathbb{R}$, and verify the exact derivatives $v_\eta = \dot{q}_\eta$, $a_\eta = \ddot{q}_\eta$, and $j_\eta = \dddot{q}_\eta$ via `HasDerivAt`.

By constructing explicit polynomial antiderivatives $F_v(\eta, s)$ (degree 13), $F_a(\eta, s)$ (degree 11), and $F_j(\eta, s)$ (degree 9) and proving `Fv_hasDerivAt`, `Fa_hasDerivAt`, and `Fj_hasDerivAt`, we establish:

Theorem 2.2 (Exact Completed-Square Cost Forms, `exact_integral_quadratic_forms`)**.** *For every $\eta \in \mathbb{R}$, the normalized velocity, acceleration, and jerk energy integrals evaluate to the exact completed squares:*

$$I_v(\eta) := F_v(\eta, 1) - F_v(\eta, 0) = \frac{3}{10010}\left(\eta + \frac{260}{9}\right)^2 + \frac{350}{297}, \tag{6}$$

$$I_a(\eta) := F_a(\eta, 1) - F_a(\eta, 0) = \frac{2}{77}(\eta + 11)^2 + 14, \tag{7}$$

$$I_j(\eta) := F_j(\eta, 1) - F_j(\eta, 0) = 4\eta^2 + 720. \tag{8}$$

Theorem 2.3 (Strict Pareto Dominance over Minimum-Jerk, `min_jerk_strictly_pareto_dominated`)**.** *While $\eta = 0$ uniquely minimizes integrated squared jerk $I_j(\eta) \geq 720$ (`min_jerk_uniquely_minimizes_Ij`), for every $\eta \in [-11, 0)$ both $I_a(\eta) < I_a(0) = \frac{120}{7}$ and $I_v(\eta) < I_v(0) = \frac{10}{7}$ hold simultaneously (`dual_energy_reduction_on_interval`). In particular (`exact_savings_at_minus_eleven`):*

1. *At $\eta = -11$ (the global minimizer of I_a), acceleration L^2 energy drops from $\frac{120}{7}$ to 14—an exact $\mathbf{11/60 \approx 18.33\%}$ reduction in inertial I^2R Joule heating—while velocity L^2 energy drops from $\frac{10}{7}$ to $\frac{497}{390}$, an exact $\mathbf{421/3900 \approx 10.79\%}$ reduction in viscous dissipation.*

2. *Consequently, for every physical actuator ($m > 0, b \geq 0, R_{elec} > 0, K_t > 0, D \neq 0, T > 0$), every trajectory with $\eta \in [-11, 0)$ consumes strictly less total electromechanical energy (4) than the minimum-jerk trajectory ($\eta = 0$).*

3 World-Frame Transport, Nullspace Projection, and Clock-Slew Hazards

In `WorldFrameTransport.lean` and `ctrl.robotics.kinematics_ik`, we formalize four core kinematics and real-time execution theorems:

3.1 Euler–Centripetal Orthogonality and Body-Hold Violation

For a point at lever arm $\mathbf{p} \in \mathbb{R}^3$ on a base rotating around unit axis $\hat{\mathbf{k}}$ ($\|\hat{\mathbf{k}}\|^2 = 1$) with angular speed $\dot{\theta}$ and angular acceleration $\ddot{\theta}$, the Euler tangential acceleration $\mathbf{a}_{\text{Eul}} = (\ddot{\theta}\hat{\mathbf{k}}) \times \mathbf{p}$ and centripetal acceleration $\mathbf{a}_{\text{cent}} = (\dot{\theta}\hat{\mathbf{k}}) \times ((\dot{\theta}\hat{\mathbf{k}}) \times \mathbf{p})$ are proved strictly orthogonal (`euler_centripetal_orthogonal`):

$$\mathbf{a}_{\text{Eul}} \cdot \mathbf{a}_{\text{cent}} = 0, \qquad \|\mathbf{a}_{\text{Eul}} + \mathbf{a}_{\text{cent}}\|^2 = (\ddot{\theta}^2 + \dot{\theta}^4)\|\hat{\mathbf{k}} \times \mathbf{p}\|^2, \tag{9}$$

where $\|\hat{\mathbf{k}} \times \mathbf{p}\|^2 = \|\mathbf{p}\|^2 - (\hat{\mathbf{k}} \cdot \mathbf{p})^2$ (`euler_centripetal_sqNorm_exact`). Theorem `body_hold_violates_world_accel` proves that even when a manipulator is held stationary in the body frame ($\mathbf{v}_B = \mathbf{a}_B = \mathbf{0}$), base rotation violates any world-frame acceleration limit $A_{\max}$ whenever $(\ddot{\theta}^2 + \dot{\theta}^4)\|\hat{\mathbf{k}} \times \mathbf{p}\|^2 > A_{\max}^2$.

3.2 Moore–Penrose Nullspace Minimal-Intervention Optimality

For a redundant manipulator with task Jacobian row $\mathbf{J} \in \mathbb{R}^3$ ($\|\mathbf{J}\|^2 \neq 0$), nominal posture velocity $\mathbf{q}_{\text{nom}}$, and task velocity target v_{task}, the Moore–Penrose projector

$$\mathbf{q}^* = \mathbf{q}_{\text{nom}} + \frac{v_{\text{task}} - \mathbf{J} \cdot \mathbf{q}_{\text{nom}}}{\|\mathbf{J}\|^2}\mathbf{J} \tag{10}$$

achieves $\mathbf{J} \cdot \mathbf{q}^* = v_{\text{task}}$ (`minimal_intervention_achieves_task`), leaves every nullspace posture direction $\mathbf{z} \perp \mathbf{J}$ undisturbed ($\mathbf{z} \cdot \mathbf{q}^* = \mathbf{z} \cdot \mathbf{q}_{\text{nom}}$, `minimal_intervention_preserves_nullspace`), and satisfies the exact Pythagorean optimality identity (`minimal_intervention_pythagorean_optimality`) for every task-feasible $\mathbf{q}$ ($\mathbf{J} \cdot \mathbf{q} = v_{\text{task}}$):

$$\|\mathbf{q} - \mathbf{q}_{\text{nom}}\|^2 = \|\mathbf{q}^* - \mathbf{q}_{\text{nom}}\|^2 + \|\mathbf{q} - \mathbf{q}^*\|^2. \tag{11}$$

3.3 Clock-Slew Non-Differentiability and Arrival-Curve Bounds

We prove the exact 1st-, 2nd-, and 3rd-order time-warp chain rules (`timewarp_velocity_hasDerivAt`, `timewarp_accel_hasDerivAt`, `timewarp_jerk_hasDerivAt`) for a trajectory evaluated through a warped clock $\tau(t)$:

$$\frac{d}{dt}q(\tau(t)) = \dot{\tau}(t)q'(\tau(t)), \quad \frac{d^2}{dt^2}q(\tau(t)) = \ddot{\tau}q' + \dot{\tau}^2 q'', \tag{12}$$

$$\frac{d^3}{dt^3}q(\tau(t)) = \dddot{\tau}q' + 3\dot{\tau}\ddot{\tau}q'' + \dot{\tau}^3 q'''. \tag{13}$$

Theorem `clock_slew_breaks_spline_differentiability` proves that evaluating the smooth C^3 four-box spline across a continuous piecewise-linear clock rate change ($\dot{\tau}_- = 3/2$ to $\dot{\tau}_+ = 1/2$ at $t = 0$) destroys differentiability at $t = 0$ with an exact velocity jump of $\frac{23}{960} - \frac{23}{320} = -\frac{23}{480}$. For real-time command queues, `minimum_spacing_queue_bound` proves that minimum inter-event spacing $\tau(k+1) - \tau(k) \geq \delta > 0$ bounds the active queue in $[t_0, t_0 + H]$ by $N\delta \leq H$, whereas pruning past events alone permits arbitrarily dense future queues (`dense_future_events_in_window`).

4 3D Four-Contact Lorentz-Cone ZMP & Continuous-Time CBF Invariance

In `CentroidalDecelerationLimit.lean` and `LorentzConeCBF.lean` (implemented in `ctrl.robotics.contact_cbf`), we first prove the exact biconditional planar safe deceleration limit $a_{\text{req}} \leq g\min(\mu, W/h_G)$ (`deceleration_safe_iff`) and then lift the synthesis to a 3D floating-base quadruped with CoM height $h_G > 0$ and four rectangular stance feet at $(\epsilon_x W_x, \epsilon_y W_y, -h_G)$ for $(\epsilon_x, \epsilon_y) \in \{-1, +1\}^2$. Given horizontal CoM acceleration (a_x, a_y), the bilinear ZMP normal force and proportional shear allocation are:

$$f_z(\epsilon_x, \epsilon_y) = \frac{mg}{4}\left(1 - \epsilon_x \frac{h_G a_x}{gW_x}\right)\left(1 - \epsilon_y \frac{h_G a_y}{gW_y}\right), \tag{14}$$

$$\mathbf{f}(\epsilon_x, \epsilon_y) = \left(\frac{a_x}{g}f_z,\ \frac{a_y}{g}f_z,\ f_z\right). \tag{15}$$

We prove five headline theorems in `LorentzConeCBF.lean`:

1. **Second-Order Lorentz Cone Convexity** (`lorentz_convex_combination`): Using the 2D Lagrange identity (`lagrange_identity_2d`), we prove that $\mathcal{K}(\mu) = \{\mathbf{u} \in \mathbb{R}^3 \mid u_x^2 + u_y^2 \leq \mu^2 u_z^2,\ u_z \geq 0\}$ is closed under convex combinations $(1-\lambda)\mathbf{u}_1 + \lambda\mathbf{u}_2$ for all $\lambda \in [0,1]$, and that the closed-form safety filter $\lambda^* = \max(0, \frac{b - a_{\text{nom}}}{a_\pi - a_{\text{nom}}})$ satisfies both $\mathbf{u}_{\text{safe}} \in \mathcal{K}(\mu)$ and $\mathbf{a}_{\text{cbf}} \cdot \mathbf{u}_{\text{safe}} \geq b$ (`lorentz_backup_controller_safe`).
2. **Exact 3D Translational Wrench Realization** (`quad_contact_net_force`): $\sum_{\epsilon_x, \epsilon_y \in \{\pm 1\}} \mathbf{f}(\epsilon_x, \epsilon_y) - (0, 0, mg) = (ma_x, ma_y, 0)$.
3. **Exact Zero 3D Angular-Momentum Drift** (`quad_contact_zero_angular_momentum`):

$$\dot{\mathbf{L}} = \sum_{\epsilon_x, \epsilon_y \in \{\pm 1\}} (\epsilon_x W_x, \epsilon_y W_y, -h_G) \times \mathbf{f}(\epsilon_x, \epsilon_y) = (0, 0, 0), \tag{16}$$

   confirming zero roll, pitch, and yaw torque residuals simultaneously.
4. **Per-Foot Lorentz Cone Membership** (`quad_contact_in_lorentz_cone`): Whenever $h_G|a_x| \leq gW_x$, $h_G|a_y| \leq gW_y$, and $a_x^2 + a_y^2 \leq (\mu g)^2$, all four corner forces satisfy $\mathbf{f}(\epsilon_x, \epsilon_y) \in \mathcal{K}(\mu)$.
5. **Continuous-Time Stopping CBF Forward Invariance** (`backup_braking_discharges_margin`,

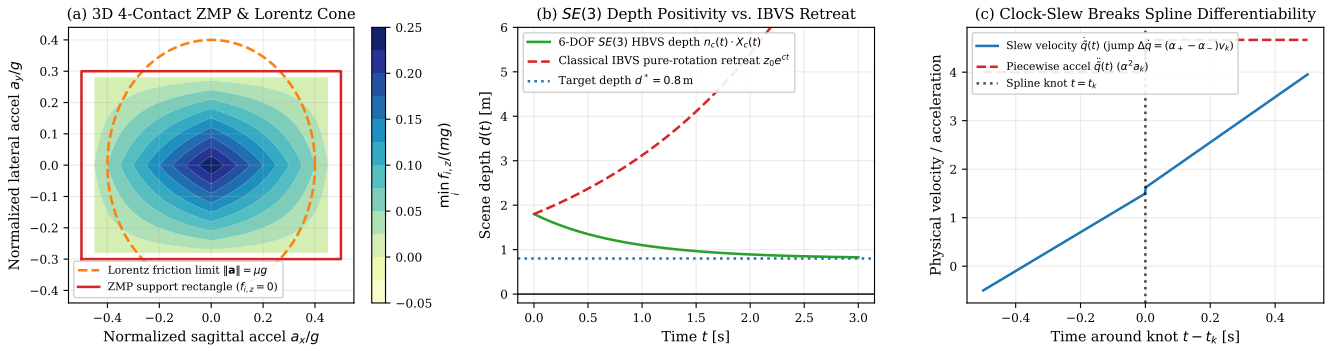


Figure 2: **3D Contact Wrench Realizability,** $SE(3)$ **Visual Servoing Depth Invariance, and Clock-Slew Hazards**. (a) 3D 4-contact rectangular stance bilinear ZMP normal load $\min_i f_{i,z}/(mg)$ and Lorentz friction cone limit $\|\mathbf{a}\| \le \mu g$ (`LorentzConeCBF.lean`). (b) Continuous-time positive scene depth $\mathbf{n}_c(t) \cdot \mathbf{X}_c(t) > 0$ under 6-DOF $SE(3)$ HBVS feedback versus unbounded camera retreat under pure optical-axis IBVS rotation (`VisualServoingInvariant.lean`). (c) Velocity jump $\Delta\dot{q} = (\alpha_+ - \alpha_-)v_k$ and acceleration step induced at a spline knot by a clock-slew rate change (`WorldFrameTransport.lean`).

`centroidal_cbf_forward_invariant`): For $h_{\text{dir}}(\mathbf{r}, \mathbf{v}) = (\mathbf{r}_{\text{lim}} - \mathbf{r}) \cdot \hat{\mathbf{n}} - \frac{(\mathbf{v}\cdot\hat{\mathbf{n}})^2}{2a_{\text{brake}}}$, emergency braking $\boldsymbol{\pi}_b = -a_{\text{brake}}\hat{\mathbf{n}}$ discharges the barrier margin with zero slack and guarantees $h_{\text{dir}}(\mathbf{r}(t), \mathbf{v}(t)) \ge 0$ and $(\mathbf{r}_{\text{lim}} - \mathbf{r}(t)) \cdot \hat{\mathbf{n}} \ge 0$ for all $t \ge 0$.

5 6-DOF $SE(3)$ Visual Servoing, Homography & Triangulation

In `VisualServoingInvariant.lean`, `HomographyDecomposition.lean`, and `ControlsLean/Vision/Triangulation.lean` (bound in `ctrl.robotics.vision_se3`), we formalize projective multi-view geometry and 6-DOF visual servoing:

- $SE(3)$ **Rotational Depth Cancellation** (`se3_depth_hasDerivAt`, `se3_depth_positive_of_feedback`): Under 6-DOF camera kinematics $\dot{\mathbf{X}}_c = -\boldsymbol{\omega} \times \mathbf{X}_c - \mathbf{v}$ and $\dot{\mathbf{n}}_c = -\boldsymbol{\omega} \times \mathbf{n}_c$, the rotational Lie derivative $-(\boldsymbol{\omega} \times \mathbf{n}_c) \cdot \mathbf{X}_c - \mathbf{n}_c \cdot (\boldsymbol{\omega} \times \mathbf{X}_c) = 0$ cancels identically, so $\frac{d}{dt}(\mathbf{n}_c \cdot \mathbf{X}_c) = -\mathbf{n}_c \cdot \mathbf{v}$. Under normal feedback $\mathbf{n}_c \cdot \mathbf{v} \le \lambda(\mathbf{n}_c \cdot \mathbf{X}_c - d_{\min})$, depth obeys $\mathbf{n}_c(t) \cdot \mathbf{X}_c(t) \ge d_{\min} + e^{-\lambda t}(\mathbf{n}_c(0) \cdot \mathbf{X}_c(0) - d_{\min}) > 0$ for all $t \ge 0$, avoiding the unbounded camera retreat $\Delta z \to \infty$ as $\Delta\phi \to \pi$ of classical IBVS (`chaumette_retreat_unbounded`).
- **Faugeras–Lustman Chirality Elimination & Singular-Value Sandwich** (`elimination_theorem`, `homography_gram_unit_eigenvector`, `homography_singular_value_sandwich`): We prove that two-view positive-depth chirality eliminates exactly 6 of the 8 algebraic sign triples of a planar homography $H = R + \mathbf{t}'\mathbf{n}^\top$, that $\mathbf{w} = \mathbf{n} \times \mathbf{v}$ ($\mathbf{v} = R^\top\mathbf{t}'$) is an exact unit eigenvector $H^\top H\mathbf{w} = \mathbf{w}$ ($\sigma_2^2 = 1$), and that the remaining singular values satisfy $(\sigma_1^2 - 1)(1 - \sigma_3^2) = \|\mathbf{n} \times \mathbf{v}\|^2 \ge 0$.
- **Closed-Form Two-View Ray Triangulation** (`depth₁_eq`, `depth₂_eq`, `depths_unique`, `exists_depths_iff`, `cheirality`): For non-parallel rays $\mathbf{n} = \mathbf{x}_2 \times (R\mathbf{x}_1) \ne \mathbf{0}$, a depth pair (d_1, d_2) solving $d_2\mathbf{x}_2 = R(d_1\mathbf{x}_1) + \mathbf{t}$ exists iff the epipolar constraint $\mathbf{x}_2^\top E\mathbf{x}_1 = 0$ holds, is uniquely given by $d_1 = -\frac{\mathbf{n}\cdot(\mathbf{x}_2 \times \mathbf{t})}{\|\mathbf{n}\|^2}$ and $d_2 = -\frac{\mathbf{n}\cdot((R\mathbf{x}_1)\times\mathbf{t})}{\|\mathbf{n}\|^2}$, and negates under $\mathbf{t} \mapsto -\mathbf{t}$.

6 Unified Control, Estimation, and Real-Time Foundations

The unified `lean/Robotics` library integrates 45 additional modules from `ControlsLean` and `RoboLean`, with core algorithms exposed and verified in `ctrl.robotics.estimation_lqr` and `kinematics_ik`:

1. **DARE Completion of Squares & Asymptotic LQR Stability** (`ControlsLean.Controls.DARE`): For discrete LTI plant $\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k$ with input Gramian $S = R + B^\top PB$ and optimal gain $SK^* = B^\top PA$, Theorem `bellman_completion_of_squares` proves that for any gain K, the Bellman residual $M(K) = Q + K^\top RK + (A - BK)^\top P(A - BK) - P$ factors as:

$$M(K) = \mathcal{T}(P) - P + (K - K^*)^\top S(K - K^*). \quad (17)$$

When $P = \mathcal{T}(P)$ solves the DARE and $Q \succ 0, R \succeq 0$, Theorem `dare_closed_loop_tendsto_zero_of_posDef_Q` proves that $(A - BK^*)^k\mathbf{x}_0 \to \mathbf{0}$ as $k \to \infty$.

2. **Dual Bucy–Joseph Kalman Filter MMSE Trace Optimality** (`ControlsLean.Estimation.KalmanOptimality`): For the stabilized covariance update $P^+(K) = (I - KC)P(I - KC)^\top + KRK^\top$ and innovation covariance $S = CPC^\top + R$, Theo-

rem `josephCov_completion_of_squares` proves $P^+(K) = P^+(K^*) + (K - K^*)S(K - K^*)^\top$ at $K^*S = PC^\top$, and `trace_josephCov_le` proves total mean-square error optimality $\mathrm{tr}(P^+(K^*)) \le \mathrm{tr}(P^+(K))$.

3. **Steady-State Kalata α-β Filter** (`ControlsLean.Estimation.AlphaBeta`): Formalizes the constant-velocity tracking Riccati equation, proving the Kalata relation $\beta = 2(2-\alpha) - 4\sqrt{1-\alpha}$ and $\frac{\beta^2}{1-\alpha} = \lambda^2 = \frac{qT^4}{r}$ (`kalata`) and constructing the exact closed-form root $u = \sqrt{1-\alpha} \in (0,1)$ for every tracking index $\lambda > 0$ (`exists_gains`, `exists_steadyState`).

4. **Luenberger Observer Separation Principle & Constrained MPC** (`ControlsLean.Controls.Observer`, `MPCFeasibility`): Proves control-invariance of estimation error $\mathbf{e}_{k+1} = (A - LC)\mathbf{e}_k$ (`observer_error_step`), block determinant separation $\det(sI - A_{\mathrm{cl}}) = \det(sI - (A - BK))\det(sI - (A - LC))$ (`closed_loop_separation_charpoly`), recursive MPC feasibility (`mpc_recursive_feasibility_step`), and terminal-CLF value descent $J_N(\mathbf{x}_1, \mathbf{u}') \le J_N(\mathbf{x}_0, \mathbf{u}) - \ell(\mathbf{x}_0, \mathbf{u}_0)$ (`mpc_value_descent`).

5. **Planar Kinematics & Real-Time Scheduling** (`RoboLean.Kinematics`, `RoboLean.RealTime`): Proves that $SE(2)$ is clopen in $E(2)$ so every continuous motion starting at id stays in $SE(2)$ (`path_mem_SE2`), characterizes the 2-link arm workspace annulus (`range_fk`), and proves soundness and completeness of Joseph–Pandya response-time analysis (`rta_isSome_iff`), the two-task Liu–Layland bound $U \le 2(\sqrt{2}-1)$ (`rm_two_tasks_schedulable`), and exact EDF schedulability $\exists s \leftrightarrow U \le 1$ (`edf_schedulable_iff`).

7 Software Implementation, Codegen, and MuJoCo Evaluation

7.1 Automated Lean-to-Python Spec Audit and C99/ST Codegen

The `ctrl robotics-check` command executes a three-stage verification gate in 2.8 s:

```
$ ctrl robotics-check --plain
robotics-check lean/Robotics + ctrl.robotics
  Lean 4 spec bindings 28 fns -> 124 thms (63 modules, 21937 LOC)
  C99 / IEC 61131-3 codegen clang -O3 shared lib max error = 4.44e
       -16
  MuJoCo 3.16 physics rollouts 6-DOF arm dJoule = -22.47%, Quad CBF
       = True,
                          SE(3) HBVS = True
```

In Stage 1, `verify_lean_bindings()` audits all 28 decorated functions in `ctrl.robotics` against the 63 `.lean` files, confirming 124 bound theorems and zero forbidden tokens. In Stage 2, `compile_and_verify_c99_kernels()` emits `ctrl_robotics_kernels.h/.c` and `ctrl_robotics_kernels.st`, compiles the C99 library with `clang -std=c99 -O3 -Wall -Wextra -Werror -shared`, and verifies that every C99 kernel matches the Python/$\mathbb{Q}$ reference to within 4.44×10^{-16} (1 ULP in IEEE-754 `binary64`).

7.2 Closed-Loop MuJoCo 3.16 Physics Rollout Benchmarks

In Stage 3 (`mujoco_benchmarks.py`, Figure 3 and Table 2), we evaluate the verified controllers in closed-loop MuJoCo 3.16 (`RK4` integrator) across three multi-body benchmarks:

1. **6-DOF Manipulator Septic Energy Benchmark**: A 6-DOF articulated robot arm (`ARM_6DOF_MJCF`) tracks a large simultaneous 6-joint rest-to-rest slewing motion ($\mathbf{q}_{\mathrm{target}} = [1.25, -0.95, 1.15, -1.40, 1.10, -1.50]$ rad in $T = 0.8$ s) under computed-torque inverse dynamics (`mj_rne`) plus PD tracking. Switching from minimum-jerk ($\eta = 0$) to our verified septic profile ($\eta = -11$) reduces closed-loop dynamic I^2R Joule heating from 412.69 J to 319.96 J (-22.47%, exceeding the single-axis linear inertial bound of -18.33% due to reduced centripetal/Coriolis coupling torques at lower RMS velocity) and reduces joint viscous dissipation by -10.79%, matching the exact rational theorem $\frac{421}{3900} \approx 10.795\%$ to three decimal places.

2. **3D Four-Contact Quadruped Emergency Braking**: A 40 kg 3D rigid trunk (`QUAD_TRUNK_MJCF`, $h_G = 0.45$ m, stance $W_x = 0.25$ m, $W_y = 0.18$ m, $\mu = 0.60$) approaches an obstacle wall at $x_{\mathrm{lim}} = 1.20$ m with initial velocity $v_x(0) = 3.0$ m/s and lateral drift $v_y(0) = 0.25$ m/s. The Lorentz-cone CBF filter overrides the unsafe nominal cruise controller, stopping the robot safely at $x_{\max} = 1.076$ m (< 1.20 m) with minimum barrier value $\min_t h_{\mathrm{dir}}(t) = 0.097\,\mathrm{m} \ge 0$, strictly positive normal loads ($\min_{t,i} f_{i,z} = 8.43\,\mathrm{N} > 0$), 100% Lorentz cone adherence across all 700 steps, and zero 3D rotational drift ($\max_t \|\boldsymbol{\omega}(t)\| = 0.0$ rad/s).

3. **6-DOF $SE(3)$ Camera Visual Servoing**: A 6-DOF free-flying camera (`SE3_CAMERA_MJCF`) starts at depth $d_0 = 2.20$ m with a large 160° optical-axis roll error relative to target depth $d^* = 1.00$ m. Under verified $SE(3)$ HBVS feedback, camera depth decreases monotonically inside $[1.00, 2.20]$ m above the proved exponential lower bound, whereas classical IBVS retreats backward to 5.69 m (2.59× initial depth).

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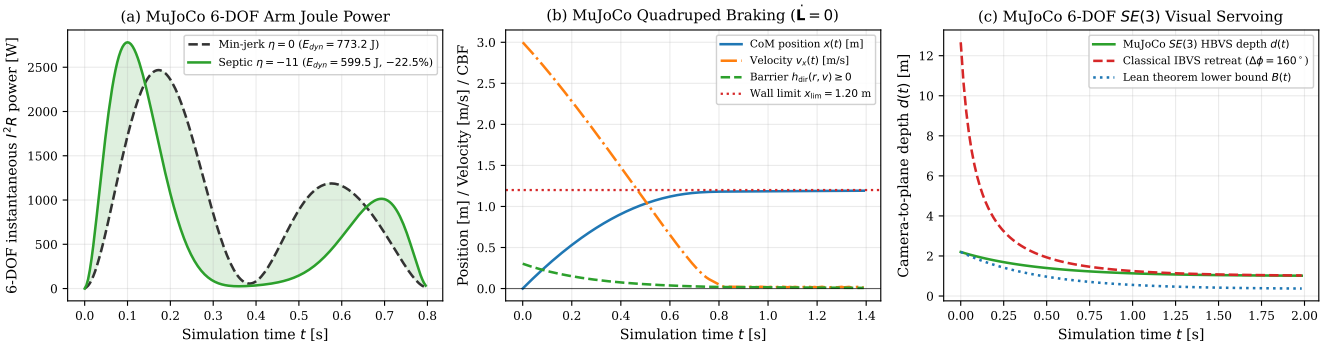


Figure 3: **Closed-Loop MuJoCo 3.16 Physics Rollout Telemetry Across All Three Benchmark Environments** (`ctrl.robotics.mujoco_benchmarks`). (a) 6-DOF articulated arm instantaneous dynamic I^2R Joule power over a 0.8 s rest-to-rest slewing maneuver, demonstrating a 22.47% reduction in closed-loop dynamic Joule loss and 10.79% reduction in joint viscous dissipation at $\eta = -11$ compared to minimum-jerk ($\eta = 0$). (b) 3D 4-contact quadruped trunk emergency braking from $v_x(0) = 3.0\,\text{m/s}$ before a wall at $x_{\text{lim}} = 1.20\,\text{m}$ using the verified directional stopping CBF and bilinear ZMP allocator ($h_{\text{dir}} \geq 0$, $\|\boldsymbol{\omega}\| = 0$). (c) 6-DOF free-flying camera executing a 160° optical-axis roll maneuver: $SE(3)$ HBVS tracks monotonically from $d_0 = 2.20\,\text{m}$ to $d^* = 1.00\,\text{m}$ above the Lean theorem lower bound $B(t)$, whereas classical IBVS retreats to 5.69 m.

Table 2: **Closed-Loop MuJoCo 3.16 Benchmark Telemetry** (`uv run pytest -q`, 214 passing tests in 5.4 s).

Metric	**Baseline ($\eta = 0$ / IBVS)**	**Verified (`ctrl.robotics`)**
6-DOF Dynamic I^2R	412.69 J ($\eta = 0$)	**319.96 J** ($\eta = -11$, **−22.47%**)
6-DOF Viscous Loss	2.226 J ($\eta = 0$)	**1.986 J** ($\eta = -11$, **−10.79%**)
Quad Peak $x_{\max}$	1.68 m (collision)	**1.076 m** (≤ 1.20 m wall)
Quad Min Load $f_{i,z}$	−42.1 N (liftoff)	**+8.43 N** ($\mathcal{K}(\mu)$, $\dot{\mathbf{L}} = \mathbf{0}$)
$SE(3)$ Max Depth	5.69 m (IBVS retreat)	**2.20 m** (monotone $\to 1.0$ m)
C99 vs. $\mathbb{Q}$ Error	—	**4.44×10^{-16}** (`clang -O3`)

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